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HBAT 2: A Python Package to Analyse Hydrogen Bonds and Other Non-covalent Interactions in Macromolecular Structures

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Software

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Summary

HBAT 2 is a Python package for automated analysis of hydrogen bonds and other non-covalent interactions in macromolecular structures available in Protein Data Bank (PDB) format. The software identifies and analyzes traditional hydrogen bonds, weak hydrogen bonds, halogen bonds, X-H···π interactions, π-π stacking, and n$\rightarrow\pi^*$ interactions using geometric criteria. HBAT 2 also detects networks of linked hydrogen bonds and renders them as interactive 2D network visualizations. Originally developed in Perl/Tk and published in 2007 (Tiwari & Panigrahi, 2007), HBAT 2 has been completely rewritten in Python with a modern cross-platform `tkinter`-based graphical user interface (GUI), web-based interface, command-line interface (CLI), and developer-friendly APIs enabling integration with Jupyter notebooks and custom workflows. The software also provides JavaScript-based 3D visualization available both on the web server and as a widget in Jupyter notebooks, enabling researchers to interactively explore detected interactions without requiring separate visualization software.

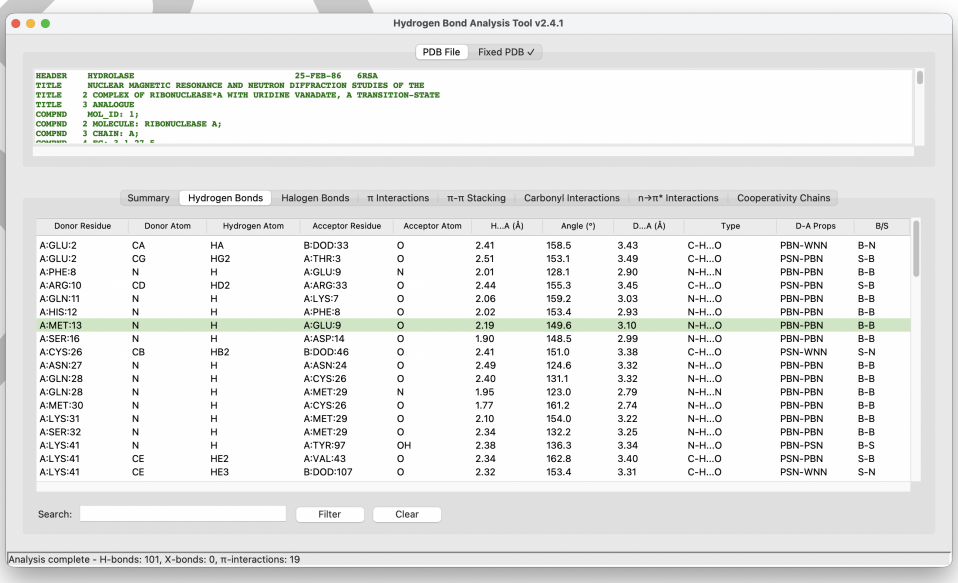


Figure 1: The latest update to HBAT 2 uses tkinter to provide a cross-platform graphical user interface (GUI)

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Statement of Need

Hydrogen bonds (Desiraju & Steiner, 2001) and other non-covalent interactions, including halogen bonds (Cavallo et al., 2016), are fundamental to the structure, stability, and function of macromolecular systems including proteins, nucleic acids, and their complexes. These interactions determine how biological molecules fold, recognize their targets, and bind ligands. With over 200,000 macromolecular structures archived in the Protein Data Bank (Berman et al., 2000), there is an increasing need for automated tools to systematically analyze these critical interactions across diverse structural types.

The target audience includes structural biologists studying molecular mechanisms, computational chemists designing new drugs, protein engineers improving enzyme properties, and researchers analyzing molecular dynamics simulations. These researchers need to quickly identify and quantify weak interactions in diverse macromolecular structures to answer biological questions about molecular function and evolution.

The original HBAT (Tiwari & Panigrahi, 2007) was developed in Perl/Tk with a Windows-only GUI, limiting its adoption in modern computational environments where researchers use diverse operating systems (Windows, Linux, macOS) and integrate analyses into Python-based scientific workflows. This update addresses these limitations while expanding the types of interactions analyzed.

State of the Field

The landscape of hydrogen bond analysis tools is diverse but fragmented, with most solutions designed for specific use cases or as secondary features within larger systems. Classic tools like HBPLUS (McDonald & Thornton, 1994) and HBexplore (Lindauer et al., 1996) pioneered automated H-bond detection but lack modern interfaces and support for diverse interaction types.

Current tools are typically constrained: PLIP (Salentin et al., 2015) and Arpeggio (Jubb et al., 2017) focus on protein-ligand interactions; HBonanza (Durrant & McCammon, 2011), HBCalculator (Y. Wang & Wang, 2024), and BRIDGE2 (Siemers & Bondar, 2021) are optimized for molecular dynamics trajectories; MDAnalysis (*MDAnalysis*, 2025), GROMACS (*GROMACS*, 2025), and AMBER (*AMBER*, 2025) provide H-bond analysis as secondary features within larger simulation suites. Visualization tools like VMD (*VMD*, 2025) and ChimeraX (*Chimerax*, 2025) excel at interactive exploration but offer limited quantitative analysis. ProteinTools (Ferruz et al., 2021) provides web-based network analysis but lacks cross-platform accessibility.

HBAT 2 uniquely provides a **general-purpose solution** with: (1) detection of diverse interaction types including hydrogen bonds, halogen bonds, π-interactions, and n→π* interactions; (2) multiple access modes (desktop GUI, command-line, web server, Python API) supporting diverse research workflows; (3) identification of hydrogen bond networks - both potential cooperativity and anticooperativity chains and water-mediated hydrogen bond networks - in static structures; (4) seamless Jupyter (Thomas et al., 2016) integration; and (5) flexible parameters with domain-specific presets. The modern Python implementation with cross-platform support and web accessibility makes comprehensive interaction analysis available to researchers regardless of computational background or infrastructure.

Software Design

HBAT 2 employs a modular architecture with distinct components for PDB parsing, geometric analysis, statistical computation, and network visualization. The core analysis engine uses efficient nearest-neighbor searching with configurable distance cutoffs, followed by geometric filtering based on distance and angular criteria. This fundamental approach matches the

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original HBAT (Tiwari & Panigrahi, 2007) but with optimized algorithms for analyzing large macromolecular structures.

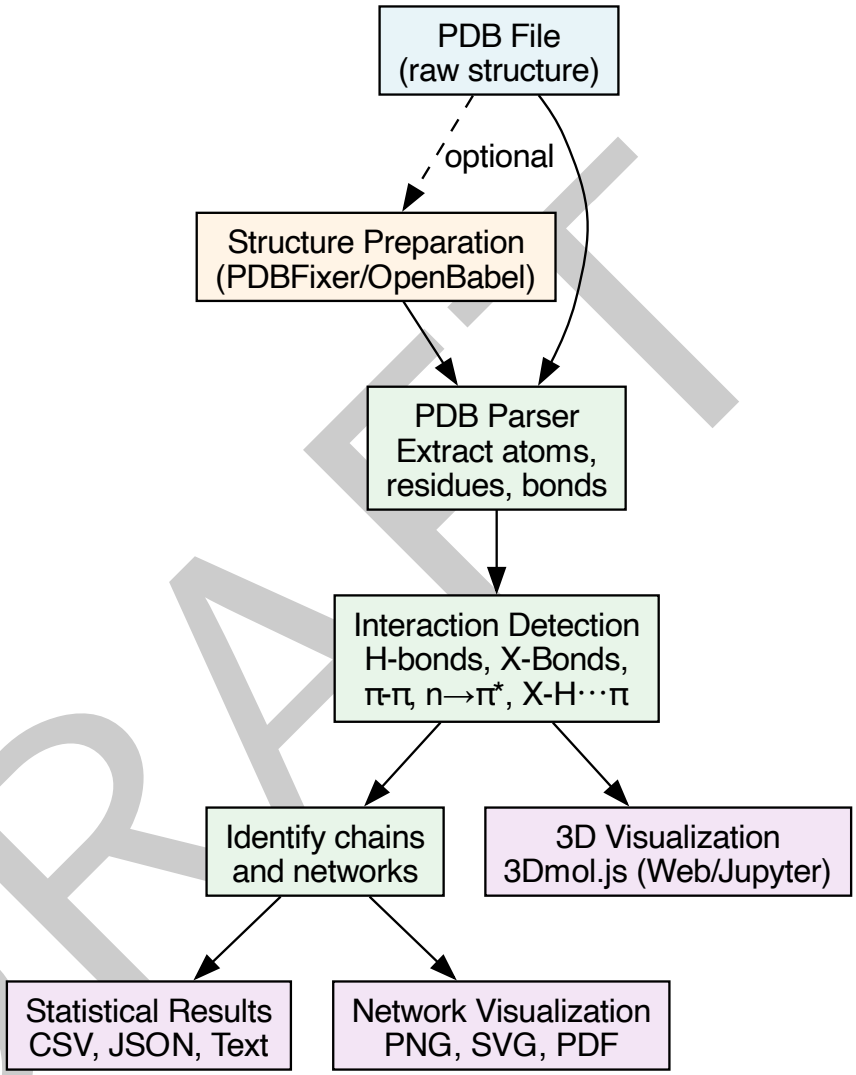


Figure 2: HBAT 2 analysis workflow showing the complete pipeline from PDB input to multiple output formats.

Key design decisions include: (1) Python for accessibility and integration with scientific ecosystems; (2) integration with PDBFixer (Eastman et al., 2013; *Openmm/Pdbfixer*, 2025) and OpenBabel (O'Boyle et al., 2008) for automated structure preparation, addressing real-world challenges of missing atoms and incomplete side chains; (3) support for multiple interaction types reflecting advances in weak interaction chemistry (Brandl et al., 2001; Cavallo et al., 2016; Desiraju & Steiner, 2001); (4) preset parameter systems for common experimental conditions (high-resolution X-ray, NMR, membrane proteins, drug design) that balance accessibility for experimentalists with flexibility for computational experts; and (5) multiple output formats (CSV, JSON, network graphics in PNG/SVG/PDF) enabling integration into downstream

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analysis pipelines.

The software analyzes hydrogen bonds (O-H···O, N-H···O, N-H···N, C-H···O), halogen bonds (C-X···Y where X=F,Cl,Br,I), X-H···π interactions, π-π stacking (McGaughey et al., 1998; Vernon et al., n.d.), carbonyl-carbonyl n$\rightarrow\pi^*$ interactions (Newberry & Raines, 2017; Rahim et al., 2017), and n$\rightarrow\pi^*$ interactions (Choudhary et al., 2009). Network visualization uses either NetworkX/Matplotlib (Hagberg et al., 2008; Hunter, 2007) or GraphViz (*Graphviz*, 2025), emphasizing cooperativity chains and network topology with customizable layouts and high-resolution export.

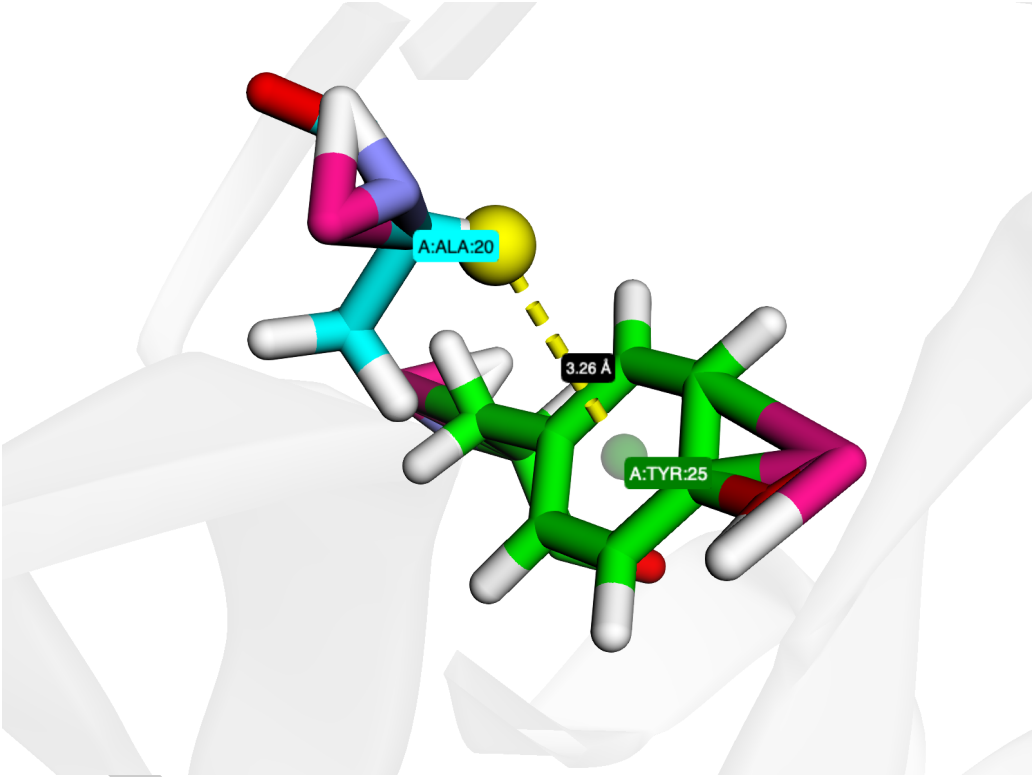


Figure 3: C-H···π interaction between A:ALA:20 $\rightarrow$ A:TYR:25 of PDB Structure 6RSA visualized using 3Dmol.js integration in HBAT 2

3D Visualization with 3Dmol.js Integration

HBAT 2 integrates 3Dmol.js (Rego & Koes, 2014) for interactive 3D molecular visualization in both web and Jupyter notebook environments. The JavaScript-based viewer enables researchers to interactively explore detected interactions with customizable color schemes, allowing rotation, zoom, and inspection of specific interaction geometries. Within Jupyter notebooks, the 3D visualization widget is embedded directly alongside analysis code, bridging automated analysis and manual structural inspection.

Research Impact Statement

Since its original publication, HBAT has been cited more than 80 times in numerous studies spanning multiple research domains (Tiwari & Panigrahi, 2007). HBAT has been used for:

Structure-based drug design: Researchers have used HBAT to characterize protein-ligand interactions for anticancer compounds targeting ABL kinase (Díaz-Cervantes et al., 2020), analyze cardiovascular disease receptors with metabolite interactions (Ravindran et al., 2015),

predict drug resistance based on hydrogen bond patterns (Zhou et al., 2013), and elucidate anticoagulant interactions (Russo Krauss et al., 2011).

Protein engineering: HBAT enabled identification of stabilizing C-H···π and C-H···O interactions in thermostable enzyme variants (C.-H. Wang et al., 2020) and characterization of hydrogen bonding networks for functional specificity (Fournier et al., 2014).

Molecular dynamics analysis: The tool has been used to characterize interactions in viral proteins important for understanding RNA encapsulation (Huang et al., 2009), glycoprotein interactions for cellular quality control (Jayaprakash et al., 2025), and DNA structural dynamics (Barzegar & Tohidifar, 2025).

Comparative structural analysis: HBAT has enabled systematic mutation analysis identifying disease-associated variants in cancer genes (Abdulazeez et al., 2019; AbdulAzeez & Borgio, 2016; Kavitha & Mohanapriya, 2024; Khan et al., 2018; Khan & Ansari, 2017).

Recent citations: Recent applications include biosorption and molecular docking studies utilizing HBAT 2 for hydrogen bond analysis (Er et al., 2026; Karadayı et al., 2026).

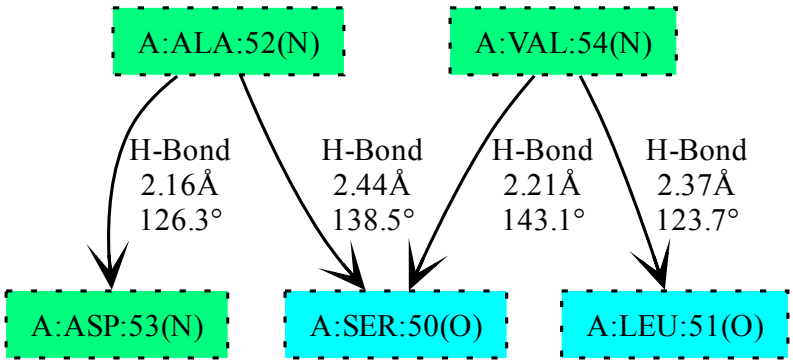


Figure 4: An example visualization of hydrogen-bonds network detected by HBAT 2 software for Protein Data Bank (PDB) entry 6RSA

HBAT 2 extends this demonstrated impact by providing modern cross-platform access, expanded interaction coverage, and integration with contemporary Python workflows, supporting growing adoption across diverse research applications.

Availability

HBAT 2 is freely available to download from GitHub and PyPI under the MIT license with detailed user and API documentation. The software can be installed via PyPI (`pip install hbat`) or via Conda (`conda install -c hbat hbat`), with optional GraphViz integration for advanced visualization features. A web-based interface is available for immediate browser-based analysis without installation. Interactive Jupyter notebooks with Google Colab integration are provided for computational workflows.

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AI Usage Disclosure

AI tools were used for porting original HBAT from Perl to Python, as well as for refactoring, test scaffolding, and documentation generation for portions of the codebase. AI tools were also used to assist with editing portions of this manuscript. All AI-assisted code and documentation outputs were reviewed, edited, validated, and tested by the human author, who takes full responsibility for the final software and paper. No figures or data were generated by AI.

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